

Solution Architecture

AI-ML & GEN-AI Track | Project Phase Documentation

Date	30 September 2026	Team ID	SWTID-2026-8540
Project Name	AI BlogNest API	Track	AI / ML / Generative AI
Team Size	5	Maximum Marks	5 Marks
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High-Level Solution Architecture

AI BlogNest API follows a layered backend architecture. A client or API tool sends requests to the Express gateway. Middleware validates tokens and permissions, service modules execute business rules, Mongoose handles database access, and AI requests are routed to Gemini-based services.



Component Description

Component	Role	Technology
Client / API Tool	Sends HTTP requests and receives JSON responses.	Web client / Postman / Thunder Client
Express Gateway	Routes requests, parses JSON, handles CORS and middleware.	Node.js + Express.js
Auth & RBAC	Validates JWT and enforces role permissions.	JWT + bcrypt.js
Service Layer	Executes blog, comment, taxonomy and AI workflows.	Node.js service modules
AI Flow	Generates content and summaries using model service integration.	Google Gemini AI
Database Layer	Persists users, posts, comments, metadata and taxonomy.	MongoDB + Mongoose